Lab Task

Question 1:

Write a C++ program using template that calculates the Euclidean norm (also known as the L2 norm or magnitude) of a vector represented by a 2D array. The Euclidean norm is defined as the square root of the sum of the squares of all elements in the vector.

Given a 2D array that represents a vector:

$$Vector = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$$

Your program should compute the Euclidean norm of the array.

Formula:

The Euclidean norm (L2 norm) of a vector $v = [v1 \quad v2 \quad \ldots \quad vn]$ is given by:

$$Norm. = \sqrt{\sum_{i=1}^{n} v_i^2}$$

For a 2D array, consider all elements as part of a single vector.

The Euclidean norm should be:

$$Norm = \sqrt{3^2 + 4^2 + 2^2 + 1^2} = \sqrt{9 + 16 + 4 + 1} = \sqrt{30}$$

Prototype: double calculateEuclideanNorm(const T array[rows][cols])

Question 2:

Write a program to keep records and perform statistical analysis for a class of students. The class may have up to 10 students. There are three quizzes during the term for computer programming. Each student is identified by a four digit roll no. The program will print the. student scores and calculate and print the following statistics.

1. High score & Low score for each quiz along with the roll number of that student.
2. Average of Each Quiz and overall average of all the 3 quizzes
3. Highest Average and Lowest average of all the three quizzes along with quiz no.
4. Highest marks & lowest marks of each student in the 3 quizzes along with the quiz no.

Prototype:

Note: You can change the prototype according to your implementation

- void **inputStudentData**(int numStudents, int numQuizzes, int rollNumbers[], int scores[][3]);
- void **printStudentScores**(int numStudents, int rollNumbers[], int scores[][3]);
- void **calculateHighLowScores**(int numStudents, int numQuizzes, int scores[][3]);

- void **calculateAverages**(int numStudents, int numQuizzes, int scores[][3]);
- void **calculateHighestLowestAverages**(int numStudents, int numQuizzes, int scores[][3]);
- void **calculateHighestLowestMarks**(int numStudents, int numQuizzes, int scores[][3]);

Question 3:

Write a C++ program using template to rotate a square matrix by 90 degrees clockwise in-place.
Given matrix:

$$Array = \begin{bmatrix} 3 & 4 & 2 \\ 5 & 6 & 1 \\ 7 & 8 & 9 \end{bmatrix}$$

After the 90 degree rotation

$$Array = \begin{bmatrix} 7 & 5 & 3 \\ 8 & 6 & 4 \\ 9 & 1 & 2 \end{bmatrix}$$

Prototype: void rotateMatrix90DegreesClockwise(T matrix[rows][cols])

Question 4:

Write a C++ program using template to traverse a 2D matrix in a spiral order and return the result as a single array. You should handle matrices of arbitrary sizes.

Example:

Given the matrix:

$$Array = \begin{bmatrix} 1 & 4 & 7 & 10 \\ 2 & 5 & 8 & 11 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

The output should be:

$$Result = [1 \quad 4 \quad 7 \quad 10 \quad 11 \quad 12 \quad 9 \quad 6 \quad 3 \quad 2 \quad 5 \quad 8]$$

Prototype: T* spiralOrder(const T matrix[rows][cols], int& size)